Impact of COVID-19 on the Labor Market in Canada

Contemporary Economic Issues: ECON 204

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Introduction

- COVID-19 profoundly disrupted Canada's labor market.
- Key points of the lecture:
  - Immediate impacts
  - Sectoral effects
  - Demographic disparities
  - Government responses
  - Long-term consequences
  - Policy implications

Immediate Impacts: Employment and Hours Worked

- **Employment losses:**
  - 3 million jobs lost (March–April 2020)
  - Unemployment peaked at 13.7% in May 2020
- **Hours worked:**
  - 28% decline in total hours worked

Case Study: Ontario's Manufacturing Sector

- 45% drop in manufacturing employment during lockdowns.
- Ford retooled plants to manufacture ventilators.

Nature of Job Losses

- Nearly 50% of job losses were in temporary or part-time positions.
- Non-essential services hit hardest (retail, hospitality, arts).

Hardest-hit Sectors

Hospitality and Tourism

- Hotel occupancy rates fell below 10% (April 2020).
- Example: Fairmont Hotels layed off over 3,000 workers.

Arts and Entertainment

- Example: Cirque du Soleil laid off 95% of its workforce.

Retail Sector

- Small businesses in Toronto's downtown core faced closures.
- E-commerce surged: Shopify reported a 76% increase in new online stores.

Resilient or Growing Sectors

- **Technology and e-commerce:**
  - Amazon hired 8,000 workers in 2020.
- **Health care:**
  - Example: British Columbia hired contact tracers, increasing capacity by 200%.
- **Essential services:**
  - Supermarkets, pharmacies, and delivery services expanded their workforce.

Gender and Youth Disparities

Gender Disparities

- Women accounted for 63% of job losses.
- Many women left the workforce due to caregiving responsibilities.

Youth Employment

- Unemployment for ages 15–24 peaked at 29.4%.
- Example: Canada Summer Jobs Program expanded to subsidize 70,000 placements.

Income Inequality

Income Inequality

- Low-wage workers faced the steepest losses.
- High-income workers in remote-capable jobs recovered quickly.

Emergency Benefits

- **CERB:** $\$2,000/month$ to 8 million Canadians.
- **CEWS:** Subsidized wages for 4 million jobs.

Sector-specific Aid

- Airlines and tourism received \$5 billion in loans.

Structural Changes and Mental Health

- **Remote Work:** Hybrid models became the norm for many sectors.
- **Automation:** Manufacturing firms accelerated automation.
- **Mental Health:** Burnout among healthcare workers rose sharply.

Policy Implications

- Strengthen social safety nets: Flexible, inclusive programs.
- Invest in skills training: Support for transitioning into high-demand fields.
- Address inequality: Subsidized childcare and equity-focused programs.
- Prepare for future shocks: Paid sick leave, workplace digitization.
- Focus on mental health: Workplace benefits and systemic reforms.

Conclusion

- The pandemic revealed vulnerabilities and opportunities in Canada's labour market.
- Collaborative policies can build a more resilient and equitable economy.

Problem 1: Dynamic Unemployment Rate Analysis

In May 2020, the labor force in Canada was 19 million people, and 2.6 million were unemployed. By September 2020, 1 million unemployed workers returned to work, but 500,000 people dropped out of the labor force.

1. Calculate the unemployment rate in May 2020.
2. Calculate the unemployment rate in September 2020, accounting for the reduced labor force.
3. By how much did the unemployment rate decrease between May and September?

Solution: Problem 1

1. Unemployment rate (May 2020):

$$\text{Unemployment Rate} = \frac{\text{Unemployed}}{\text{Labor Force}} \times 100 = \frac{2.6M}{19M} \times 100 = 13.68\%.$$

2. Reduced labor force:

$$\text{New Labor Force} = 19M - 0.5M = 18.5M.$$

Unemployed workers:

$$\text{New Unemployed} = 2.6M - 1M = 1.6M.$$

Unemployment rate (September 2020):

$$\frac{1.6M}{18.5M} \times 100 = 8.65\%.$$

3. Decrease in unemployment rate:

$$13.68\% - 8.65\% = 5.03\% \text{ (percentage points)}.$$

Problem 2: Sectoral Employment Decline and Recovery

The hospitality sector employed 1.5 million workers in January 2020. During the first wave, employment dropped by 45%. By December 2021, employment recovered by 60% of the initial job losses.

1. How many workers were employed at the peak of the job losses?
2. How many jobs were recovered by December 2021?
3. What percentage of the pre-pandemic workforce was employed by December 2021?

Solution: Problem 2

1. Jobs lost:

$$0.45 \times 1.5M = 675,000.$$

Workers remaining:

$$1.5M - 675,000 = 825,000.$$

2. Jobs recovered:

$$0.60 \times 675,000 = 405,000.$$

Workers employed:

$$825,000 + 405,000 = 1,230,000.$$

3. Percentage of pre-pandemic workforce:

$$\frac{1,230,000}{1,500,000} \times 100 = 82\%.$$

Problem 3: CERB and Household Income Analysis

The Canadian government provided CERB payments of \$2,000/month to 8 million people for 6 months. Suppose the average pre-pandemic household income for these recipients was \$3,500/month.

1. Calculate the total CERB expenditure over 6 months.
2. What percentage of their pre-pandemic income did CERB replace?
3. If household expenses averaged \$2,700/month, calculate the monthly shortfall or surplus during CERB.

Solution: Problem 3

1. Total CERB expenditure:

$$8M \times \$2,000 \times 6 = \$96 \text{ billion.}$$

2. Percentage of income replaced:

$$\frac{\$2,000}{\$3,500} \times 100 = 57.14\%.$$

3. Monthly shortfall:

$$\$2,700 - \$2,000 = \$700 \text{ (shortfall).}$$

Problem 4: Wage Subsidy Program and Business Payrolls

A company applied for CEWS to cover 75% of payroll costs for 10 employees earning \$4,000/month each. The program lasted for 6 months.

1. Calculate the total subsidy the company received.
2. If the company's revenue was \$300,000 during the 6 months, what percentage of its revenue was used to cover the remaining payroll costs?
3. Assuming the business's total operating costs (excluding payroll) were \$120,000, calculate the profit or loss during the 6-month period.

Solution: Problem 4

1. Total subsidy:

$$75\% \times (10 \times \$4,000) \times 6 = \$180,000.$$

2. Remaining payroll:

$$25\% \times (10 \times \$4,000) \times 6 = \$60,000.$$

Percentage of revenue:

$$\frac{\$60,000}{\$300,000} \times 100 = 20\%.$$

3. Profit or loss:

Revenue - Total Costs $= \$300,000 - (\$60,000 + \$120,000) = \$120,000$ p

Problem 5: E-commerce Sector Growth

The e-commerce sector grew by 76% in 2020 compared to pre-pandemic levels. In 2019, the sector generated $50 billion in revenue. By 2021, growth slowed to 12% compared to 2020.

1. Calculate the revenue in 2020.
2. Calculate the revenue in 2021.
3. What was the compound annual growth rate (CAGR) from 2019 to 2021?

Solution: Problem 5

1. Revenue in 2020:

$$50B \times (1 + 0.76) = 50B \times 1.76 = 88B.$$

2. Revenue in 2021:

$$88B \times (1 + 0.12) = 88B \times 1.12 = 98.56B.$$

3. CAGR:

$$\text{CAGR} = \left(\frac{\text{Revenue in 2021}}{\text{Revenue in 2019}}\right)^{\frac{1}{2}} - 1$$

Substituting values:

$$\text{CAGR} = \left(\frac{98.56}{50}\right)^{\frac{1}{2}} - 1 = (1.9712)^{0.5} - 1 = 1.396 - 1 = 0.396 = 39.6\%.$$

Problem 6: Regional Tourism Losses

In 2020, tourism in British Columbia generated \$10 billion less than the previous year, representing a 60% decline. Assume tourism gradually recovers, growing 20% annually starting in 2021.

1. What was the total tourism revenue in 2019?
2. Calculate the tourism revenue in 2021 and 2022.
3. How many years will it take to fully recover to 2019 levels?

Solution: Problem 6

1. Revenue in 2019:

$$\text{Revenue in 2019} = \frac{\$10B}{0.6} = 16.67B.$$

2. Revenue in 2021:

$$10B \times (1 + 0.20) = 10B \times 1.2 = 12B.$$

Revenue in 2022:

$$12B \times (1 + 0.20) = 12B \times 1.2 = 14.4B.$$

3. Time to recover:

$$10B \times (1.20)^x = 16.67B.$$

Taking the natural logarithm:

$$x = \frac{\ln(1.667)}{\ln(1.20)} \approx 2.8.$$

Full recovery will take approximately 3 years.

Problem 7: Labor Force Dynamics

In 2020, the working-age population in Canada was 30 million. The labor force participation rate was 65%, and the employment-to-population ratio was 58%.

1. Calculate the number of people in the labor force and the number employed.
2. If 500,000 workers dropped out of the labor force in 2021, calculate the new participation rate (assume the working-age population remains constant).
3. If employment increased by 800,000 in 2021, calculate the new employment-to-population ratio.

Solution: Problem 7

1. Labor force:

$$\text{Labor Force} = 0.65 \times 30M = 19.5M.$$

   Number employed:

$$\text{Employed} = 0.58 \times 30M = 17.4M.$$

2. New participation rate:

$$\text{New Labor Force} = 19.5M - 0.5M = 19M.$$

$$\text{Participation Rate} = \frac{19M}{30M} \times 100 = 63.33\%.$$

3. New employment-to-population ratio:

$$\text{New Employment} = 17.4M + 0.8M = 18.2M.$$

$$\text{Employment-to-Population Ratio} = \frac{18.2M}{30M} \times 100 = 60.67\%.$$